PRATEIK SINHA

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EDUCATION

Carnegie Mellon University *Aug 2025 – Present | Expected Graduation: Dec 2026*
M.S. Machine Learning, School of Computer Science, GPA: 4.0/4.0

University of California, Los Angeles (UCLA) *2020 – 2024*
B.S. Double major in Mathematics of Computation & Statistics and Data Science, GPA: 3.901/4.0

PUBLICATIONS & PATENTS [Google Scholar]

- **Seizure-Semiology-Suite (S^3): A Clinically Multimodal Dataset, Benchmark, and Models for Seizure Semiology Understanding**
  Lina Zhang, Jiarui Cui, Tonmoy Monsoor, Peizheng Li, Xinyi Peng, Chong Han, *Prateik Sinha* et al. | ICML 2026 (Spotlight)
- **Can Multimodal Large Language Models Understand Pathologic Movements? A Pilot Study On Seizure Semiology**
  Lina Zhang, Tonmoy Monsoor, *Prateik Sinha* et al. | IEEE EMBC 2026.
- **Interictal Mini-Seizures: Recurrent Neuronal Synchronization Events Driven by the Epileptogenic Zone** [Paper]
  Tonmoy Monsoor, Sotaro Kanai, *Prateik Sinha* , et al. | medRxiv preprint
- **Time-Resolved Neuronal Network Dynamics Distinguish Pathological States in Organoid Models** [Paper]
  Colin M. McCrimmon*, *Prateik Sinha** , Qing Cao*, Tonmoy Monsoor*, et al. | ICASSP 2026.
- **Automated Seizure Classification Using Multimodal Large Language Models** [Paper]
  Lina Zhang, Richard Jiang, Tonmoy Monsoor, Jessica Nichole Pasqua, Colin M. McCrimmon, *Prateik Sinha* , et al. | medRxiv preprint.
- **AtmosArena: Benchmarking Foundation Models for Atmospheric Sciences** [Paper] [Code]
  Tung Nguyen, *Prateik Sinha* , Advit Deepak, Karen A. McKinnon, Aditya Grover. | NeurIPS 2024 Workshops: FM4Science, CCAI.
- **Methods And Systems for Automatically Creating Engineering Drawings**
  Sinha, Prateik & Savant, Shrikant. 2025. US Application No. 63/837,442, filed July 2, 2025. Provisional Patent.

EXPERIENCE

Research Intern | Sakana AI *May 2026–Sep 2026*
- Adapting Sakana AI's Continuous Thought Machine neural network architecture for ARC-AGI 3-style puzzles.

Machine Learning Engineer, SolidWorks | Dassault Systèmes *Jul–Sep 2023 (Intern) | Apr 2024–Aug 2025 (Full Time)*
- Developed and patented a machine learning method to automatically generate engineering drawings from CAD models. Productionized the method and deployed it on SolidWorks Desktop and xDesign using Python, JavaScript and C++.
- Built ML models to generate and complete simple 3D models, extract specifications of 3D objects from 2D images, and detect features in generic 3D files.

Data Science Intern | Zelis Healthcare *Jun 2022 – Dec 2022*
- Built a pipeline and model to re-price insurance claims based on historical data with Python, R, Snowflake, and Microsoft Azure.
- Developed and maintained dashboards to track insurances claims, their re-pricings, and appeal status. Web-scraped new data using Python (Selenium) and automated cleaning and transforming incoming raw data, as well as updating tables in Snowflake and Azure.

RESEARCH

Carnegie Mellon University | Prof. Andrej Risteski *Aug 2025 – Present*
Topic: Synthetic data generation for transition path sampling in molecular dynamics.

UCLA | Prof. Aditya Grover *Jun 2023 – Mar 2024*
Topic: Fine-tuning and benchmarking foundation models for atmospheric sciences; consolidating station-level data with the ERA5 dataset.

UCLA | Prof. Vwani Roychowdhury *Apr 2023 – Mar 2024*
Topic: Using network dynamics and machine learning to study epilepsy and human brain organoids; VLMs for seizure semiology.

PROJECTS

LA Hacks 2023: 3rd place out of 187 teams *April 2023*
- Developed an app, people2vec, which matches like-minded people based on their personality by generating an embedding of their YouTube watch history and performing a similarity search against other users. Users can also explore an interactive 3D graph of how their tastes align with their matches.

HackMIT 2022: 2nd place in 'Best Use of Blockchain for Social Good' category *Oct 2022*
- Built a crowd-sourced knowledge database, WikiSafe, which stakes all changes/edits to articles on the Ethereum blockchain to create a permanent, immutable record. Also summarizes content, captions images for accessibility, and generates relevant images. Made using React, Flask, Solidity and Web3.js.

SKILLS

Machine Learning Research/Engineering, AI for Science, Computer Vision (2D/3D), NLP, Data Science, Statistical Analysis, Software Development & Engineering
Languages & Tools: Python (PyTorch, TensorFlow, JAX, NumPy), C++, C, JavaScript (React), TypeScript, R, SQL, Snowflake, Azure, AWS, Tableau